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Aircraft pitch control system with electronically geared elevator

Abstract

Aircraft pitch control systems and methods are disclosed. In one exemplary embodiment, an aircraft pitch control system comprises a movable horizontal stabilizer and an elevator movably coupled to the horizontal stabilizer. The elevator is electronically geared to the horizontal stabilizer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) The present application is a continuation of U.S. patent application Ser. No. 16/095,831 filed on Apr. 19, 2017, which is a national phase application under 35 U.S.C. 371 of international patent application No. PCT/IB2017/052257 filed on Apr. 19, 2017, which claims priority to U.S. provisional patent application No. 62/326,971 filed on Apr. 25, 2016, the entire contents of all of which are hereby incorporated by reference.

TECHNICAL FIELD
(1) The disclosure relates generally to pitch control systems of aircraft, and more particularly to aircraft pitch control systems with electronically geared elevators.
BACKGROUND OF THE ART
(2) The horizontal stabilizer on an aircraft is typically a movable aerodynamic surface located in an aft portion of the aircraft and that is used to provide longitudinal (pitch) stability and control for the aircraft. In order to set the pitch angle of the aircraft, the horizontal stabilizer can be trimmed (actuated) such that it selectively produces positive or negative lift, or is neutral. Conventional horizontal stabilizers can be moved about an axis running parallel to the aircraft lateral axis, so that they can be trimmed over a predetermined angular range. The magnitude and direction of the aerodynamic force (i.e., positive lift or negative lift) that is produced can be varied by varying the incidence angle of the horizontal stabilizer.
(3) A horizontal stabilizer must be sufficiently large to produce the desired aerodynamic effect at various operating conditions (e.g., flap configurations, speeds, etc.) of an aircraft. The size of the horizontal stabilizer of an aircraft affects the amount of drag that it produces during flight and also affects the overall weight of the aircraft.
SUMMARY
(4) In one aspect, the disclosure describes an aircraft pitch control system. The system comprises: a first actuator configured to actuate a movable stabilizer associated with a pitch of the aircraft; a second actuator configured to actuate an elevator movably coupled to the stabilizer; and one or more data processors operatively coupled to the first actuator and to the second actuator; and machine-readable memory storing instructions executable by the one or more data processors and configured to cause the one or more data processors to: using data representative of a commanded

stabilizer actuation amount and data representative of a gearing relationship between the stabilizer and the elevator, determine a corresponding elevator actuation amount associated with the commanded stabilizer actuation amount; and generate an output for causing the first actuator to actuate the stabilizer according to the commanded stabilizer actuation amount and for causing the second actuator to actuate the elevator according to the corresponding elevator actuation amount.
(5) The gearing relationship may comprise: a first portion over a first range of stabilizer actuation amounts in which a first non-zero stabilizer actuation amount requires a corresponding first non-zero elevator actuation amount; and a second portion over a second range of stabilizer actuation amounts in which a second non-zero stabilizer actuation amount requires no elevator actuation amount.
(6) The second range of stabilizer actuation amounts of the second portion may comprise a neutral position of the stabilizer.
(7) The gearing relationship may comprise a third portion over a third range of stabilizer actuation amounts in which a third non-zero stabilizer actuation amount requires a corresponding second non-zero elevator actuation amount.
(8) The second range of stabilizer actuation amounts may be between the first range of stabilizer actuation amounts and the third range of stabilizer actuation amounts.
(9) In some embodiments, at least one of the first portion and third portion of the gearing relationship is linear.
(10) In some embodiments, each of the first portion and third portion of the gearing relationship is linear.
(11) The gearing relationship may comprise a dead zone in which stabilizer actuation requires no elevator actuation. The dead zone may include a neutral position of the stabilizer.
(12) The gearing relationship may comprise a first portion over a first range of stabilizer actuation amounts and a second portion over a second range of stabilizer actuation amounts where the first portion and the second portion have different slopes.
(13) The gearing relationship may comprise a linear portion.
(14) The gearing relationship may comprise a non-linear portion.
(15) The instructions may be configured to cause the one or more data processors to reset a neutral position of the elevator to the corresponding elevator actuation amount.
(16) The instructions may be configured to cause the one or more data processors to change the gearing relationship based on one or more operating parameters of the aircraft.
(17) In another aspect, the disclosure describes an aircraft pitch control system comprising: a movable horizontal stabilizer; and an elevator movably coupled to the horizontal stabilizer, the elevator being electronically geared to the horizontal stabilizer.
(18) The elevator may be electronically geared to the horizontal stabilizer according to a gearing relationship comprising: a first portion over a first range of stabilizer actuation amounts in which a first non-zero stabilizer actuation amount requires a corresponding first non-zero elevator actuation amount; and a second portion over a second range of stabilizer actuation amounts in which a second non-zero stabilizer actuation amount requires no elevator actuation amount.
(19) The second range of stabilizer actuation amounts of the second portion may comprise a neutral position of the stabilizer.
(20) The gearing relationship may comprise a third portion over a third range of stabilizer actuation amounts in which a third non-zero stabilizer actuation amount requires a corresponding second non-zero elevator actuation amount.
(21) The second range of stabilizer actuation amounts may be between the first range of stabilizer actuation amounts and the third range of stabilizer actuation amounts.
(22) In some embodiments, at least one of the first portion and third portion of the gearing relationship is linear.
(23) In some embodiments, each of the first portion and third portion of the gearing relationship is

linear.

(24) The elevator may be electronically geared to the horizontal stabilizer according to a gearing relationship comprising a dead zone in which stabilizer actuation requires no elevator actuation. The dead zone may include a neutral position of the stabilizer.

(25) The elevator may be electronically geared to the horizontal stabilizer according to a gearing relationship comprising a first portion over a first range of stabilizer actuation amounts and a second portion over a second range of stabilizer actuation amounts where the first portion and the second portion have different slopes.

(26) The gearing relationship may comprise a linear portion.

(27) The gearing relationship may comprise a non-linear portion.

(28) The gearing relationship may be changeable based on one or more operating parameters of the aircraft.

(29) In another aspect, the disclosure describes are aircraft comprising a pitch control system as disclosed herein.

(30) In a further aspect, the disclosure describes a method for controlling the pitch of an aircraft using a movable stabilizer of the aircraft and an elevator movably coupled to the stabilizer. The method comprises: actuating the stabilizer; and actuating the elevator based on an electronic gearing relationship between the stabilizer and the elevator.

(31) The gearing relationship may comprise: a first portion over a first range of stabilizer actuation amounts in which a first non-zero stabilizer actuation amount requires a corresponding first non-zero elevator actuation amount; and

(32) a second portion over a second range of stabilizer actuation amounts in which a second non-zero stabilizer actuation amount requires no elevator actuation amount.

(33) The second range of stabilizer actuation amounts of the second portion may comprise a neutral position of the stabilizer.

(34) The gearing relationship may comprise a third portion over a third range of stabilizer actuation amounts in which a third non-zero stabilizer actuation amount requires a corresponding second non-zero elevator actuation amount.

(35) The second range of stabilizer actuation amounts may be between the first range of stabilizer actuation amounts and the third range of stabilizer actuation amounts.

(36) In some embodiments, at least one of the first portion and third portion of the gearing relationship is linear.

(37) In some embodiments, each of the first portion and third portion of the gearing relationship is linear.

(38) The gearing relationship may comprise a dead zone in which stabilizer actuation requires no elevator actuation. The dead zone may include a neutral position of the stabilizer.

(39) The gearing relationship may comprise a first portion over a first range of stabilizer actuation amounts and a second portion over a second range of stabilizer actuation amounts where the first portion and the second portion have different slopes.

(40) The gearing relationship may comprise a linear portion.

(41) The gearing relationship may comprise a non-linear portion.

(42) The method may comprise resetting a neutral position of the elevator to an elevator actuation amount attributed to the electronic gearing relationship.

(43) The method may comprise changing the gearing relationship based on one or more operating parameters of the aircraft.

(44) Further details of these and other aspects of the subject matter of this application will be apparent from the detailed description and drawings included below.

Description

DESCRIPTION OF THE DRAWINGS

(1) Reference is now made to the accompanying drawings, in which:

(2) FIG. **1** is a top plan view of an exemplary aircraft comprising a pitch control system as disclosed herein;

(3) FIG. **2** is a schematic representation of an exemplary pitch control system as disclosed herein;

(4) FIG. **3** is a graphical representation of three exemplary gearing relationships for the pitch control system of FIG. **2**;

(5) FIG. **4** is a schematic illustration of a horizontal stabilizer and an elevator of the aircraft of FIG. **1** where the horizontal stabilizer is actuated to a position away from its neutral position and the elevator is actuated relative to the horizontal stabilizer according to an exemplary gearing relationship; and

(6) FIG. **5** is a flowchart illustrating a method for controlling the pitch of an aircraft.

DETAILED DESCRIPTION

(7) The present disclosure relates to aircraft pitch control systems with electronically geared elevators. In some embodiments, the use of one or more elevators that are geared to the movement of an associated horizontal stabilizer (i.e., tailplane) may improve the effectiveness of the stabilizer. Accordingly, in some embodiments, the use of a geared elevator may permit the use of a stabilizer of a smaller size and hence reduced weight compared to stabilizers that have non-geared elevators movably coupled thereto.

(8) In some embodiments, the use of electronic gearing may provide flexibility with making changes to an associated gearing relationship compared to mechanical motion transfer means (e.g., gears, linkages) between the stabilizer and the elevator. In some embodiments, the use of electronic gearing may also allow the use of different gearing relationships at different times depending on the operating condition of an aircraft. Such flexibility, may be advantageous for example during the development of an aircraft and during flight testing where adjustments to the gearing relationship may be required. The use of electronic gearing may provide operational advantages such as permitting the use of complex gearing relationships that would otherwise be difficult to achieve using mechanical motion transfer means.

(9) Aspects of various embodiments are described through reference to the drawings.

(10) FIG. **1** is a top plan view of an exemplary aircraft **10** which may comprise a pitch control system and perform associated methods disclosed herein. Aircraft **10** may be any type of aircraft such as corporate (e.g., business jet), private, commercial and passenger aircraft suitable for civil aviation. For example, aircraft **10** may be a narrow-body, twin-engine jet airliner. Aircraft **10** may be a fixed-wing aircraft.

(11) Aircraft **10** may comprise one or more wings **12** including one or more flight control surfaces **14**, fuselage **16**, one or more engines **18** and empennage **20** of known or other type. One or more of engines **18** may be mounted to fuselage **16**. Alternatively, or in addition, one or more of engines **18** may be mounted to wings **12**.

(12) Empennage **20** may comprise vertical stabilizer **22** to which a movable rudder may be coupled and used to impart a turning or yawing motion to aircraft **10**. Empennage **20** may also comprise horizontal stabilizer **24** (referred hereinafter as “stabilizer **24**”) and one or more elevators **26**R and **26**L (referred generically as “elevator **26**”) movably coupled to stabilizer **24**. Stabilizer **24**, sometimes called “tailplane”, may extend substantially symmetrically and laterally from each side of vertical stabilizer **22** where right elevator **26**R may be disposed on a right hand side of vertical stabilizer **22** and left elevator **26**L may be disposed on a left hand side of vertical stabilizer **22**. Stabilizer **24** may be movably coupled to empennage **20** so as to be “trimmable”. Stabilizer **24** and elevators **26**A and **26**B may be independently controllably actuatable and be part of a pitch control system of aircraft **10**. It is understood that stabilizer **24** may not necessarily be absolutely horizontal but may instead be at any orientation suitable for providing pitch stability or control.

(13) FIG. **2** is a schematic representation of an exemplary pitch control system **28** of aircraft **10** as disclosed herein. Pitch control system **28** may permit electronic gearing between stabilizer **24** and elevator **26** so that a commanded actuation of stabilizer **24** may automatically result in a corresponding actuation of elevator **26** at least in some range of movement of stabilizer **24**. This may increase the effectiveness (e.g., power) of stabilizer **24** during operation (e.g., flight, cruise, take-off) of aircraft **10** compared to a conventional stabilizer of similar size not equipped with a geared elevator. In some embodiments, the use of geared elevator **26** may permit the reduction of the total range of travel of stabilizer **24** and hence reduce the size of the actuation mechanism(s) required for stabilizer **24** compared to stabilizers that have non-geared elevators movably coupled thereto. Accordingly, the use of geared elevator **26** may permit stabilizer **24** to be of reduced cost, size and/or weight compared to a conventional stabilizer of similar performance but not equipped with a geared elevator.
(14) In some embodiments, the effectiveness of stabilizer **24** due to electronic gearing may be increased at least in part due to camber being generated in the horizontal tail (stabilizer **24**+elevator **26**) in areas of the flight envelope where relatively high (positive or negative) tail lift is required.
(15) The use of electronic gearing between stabilizer **24** and elevator **26** may permit synchronized actuation of stabilizer **24** and elevator **26** where elevator **26** may be actuated as a function of (e.g., in proportion to) stabilizer **24** actuation without the use of mechanical motion transfer means to achieve such synchronized actuation. In other words, stabilizer **24** may be considered a master device and elevator **26** may be considered a slave device where the master device and the slave device are actuated by independently controlled actuators. As the actuation of the slave device follows the actuation of the master device according to a relationship (e.g., constant ratio), the effect achieved may be similar to that of two devices that are mechanically geared. The synchronized actuation of stabilizer **24** and elevator **26** using such electronic gearing may be controlled by suitable (e.g., position) sensing and feedback control methods of known or other types.
(16) Electronic gearing can have advantages over the use of mechanical motion transfer means (e.g., gears, linkages). One notable advantage is flexibility because the gearing relationship (i.e., gear ratio) can be changed without having to redesign/build a mechanical motion transfer means. This can be advantageous during (e.g., flight, wind tunnel) testing of aircraft **10** for example where some adjustments to the gearing relationship may be required. Another advantage is that the nature of the gearing relationship is not dictated by limitations, complexities and weight associated with mechanical motion transfer means.
(17) As shown schematically in FIG. **2**, pitch control system **28**, may in some embodiments, comprise elevator actuators **30**R and **30**L (referred generically as “elevator actuator **30**”) respectively associated with elevators **26**R and **26**L. In some embodiments of aircraft **10**, elevator actuators **30**R and **30**L may be used to separately actuate right elevator **26**R and left elevator **26**L. However, in some embodiments, right elevator **26**R and left elevator **26**L may be actuated together as a unit by one or more common actuators **30**. In various embodiments, right elevator **26**R and left elevator **26**L may be separately actuatable or may be connected together so as to be actuatable as a single flight control surface via one or more actuators **30**. Pitch control system **28** may also comprise one or more stabilizer actuators **32** configured to actuate movable stabilizer **24** of aircraft **10**.
(18) FIG. **2** shows elevators **26**R and **26**L as being separate from stabilizer **24** but it is understood that elevators **26**R and **26**L may be movably coupled (e.g., hinged) to stabilizer **24**. Elevator **26** may be actuated/deflected by some amount measured as angular displacement relative to an elevator neutral position. Stabilizer **24** may also be actuated/deflected by some amount measured as angular displacement relative to a stabilizer neutral position.
(19) Pitch control system **28** may be disposed onboard of aircraft **10** and may comprise one or more computers **34** (referred hereinafter in the singular) operatively coupled to elevator actuator **30** and

to stabilizer actuator **32**. It is understood that computer **34** may be directly or indirectly (e.g., via intermediate device(s)) coupled to elevator actuator **30** and stabilizer actuator **32** so as to impart some control over the operation of elevator actuator **30** and stabilizer actuator **32**. Computer **34** may comprise one or more data processors **36** (referred hereinafter in the singular) of known or other type and which may be used to perform methods disclosed herein in entirety or in part. In some embodiments, methods disclosed herein may be performed using a single data processor **36** or, alternatively, parts of the methods disclosed herein could be performed using multiple data processors **36**. Computer **34** may comprise machine-readable memory **38** storing instructions **40** executable by data processor **36** and configured to cause data processor **36** to carry out one or more tasks associated with controlling the pitch of aircraft **10** via elevator actuator **30** and stabilizer actuator **32**.

(20) For example, computer **34** may receive input(s) **42** in the form of data or information that may be processed by data processor **36** based on instructions **40** in order to generate output **44**. For example, input **42** may comprise information (data) representative of a command associated with the pitch of aircraft **10**. Input **42** may comprise one or more signals representative of an input received from a pilot of aircraft **10** via input device **46** for example. Input device **46** may be of the type known as “side stick” or “control yoke” typically used by a pilot to input pitch commands. Alternatively, input **42** may be provided by another computer or control system (e.g., auto-flight, auto-trim) of aircraft **10**. Alternatively, input **42** could also be produced/derived within computer **34** and subsequently used by data processor **36**. Input **42** may be representative of a commanded actuation amount for stabilizer **24**.

(21) Computer **34** may be part of an avionics suite of aircraft **10**. For example, in some embodiments, computer **34** may carry out additional functions than those described herein. In some embodiments, computer **34** may be of the type known as a primary flight control computer (PFCC) of aircraft **10**. In some embodiments, pitch control system **28** may be part of a fly-by-wire control system of known or other type for aircraft **10**.

(22) Data processor **36** may comprise any suitable device(s) configured to cause a series of steps to be performed by computer **34** so as to implement a computer-implemented process such that instructions **40**, when executed by computer **34**, may cause the functions/acts specified in the methods described herein to be executed. Data processor **36** may comprise, for example, any type of general-purpose microprocessor or microcontroller, a digital signal processing (DSP) processor, an integrated circuit, a field programmable gate array (FPGA), a reconfigurable processor, other suitably programmed or programmable logic circuits, or any combination thereof.

(23) Memory **38** may comprise any suitable known or other machine-readable storage medium. Memory **38** may comprise non-transitory computer readable storage medium such as, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. Memory **38** may include a suitable combination of any type of computer memory that is located either internally or externally to computer **34**. Memory **38** may comprise any storage means (e.g. devices) suitable for retrievably storing machine-readable instructions **40** executable by data processor **36**.

(24) Various aspects of the present disclosure nay be embodied as systems, devices, methods and/or computer program products. Accordingly, aspects of the present disclosure may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects. Furthermore, aspects of the present disclosure may take the form of a computer program product embodied in one or more non-transitory computer readable medium(ia) (e.g., memory **38**) having computer readable program code (e.g., instructions **40**) embodied thereon. The computer program product may, for example, be executed by computer **34** to cause the execution of one or more methods disclosed herein in entirety or in part.

(25) Computer program code for carrying out operations for aspects of the present disclosure in

accordance with instructions **40** may be written in any combination of one or more programming languages, including an object oriented programming language such as Java, Smalltalk, C++ or the like and conventional procedural programming languages, such as the "C" programming language or other programming languages. Such, program code may be executed entirely or in part by computer **34** or other data processing device(s). It is understood that, based on the present disclosure, one skilled in the relevant arts could readily write computer program code for implementing the methods disclosed herein.

(26) In various embodiments, instructions **40** may be configured to cause data processor **36** to use data representative of a commanded stabilizer actuation amount (e.g., input **42**) and data representative of one or more gearing relationships **48** between stabilizer **24** and elevator **26** to determine a corresponding elevator actuation amount associated with the commanded stabilizer actuation amount. Instructions **40** may further be configured to cause computer **34** to generate an output **44** (e.g., signals, data) for causing actuator **32** to actuate stabilizer **24** according to the commanded stabilizer actuation amount and for causing actuator **30** to actuate elevator **26** according to the corresponding elevator actuation amount.

(27) FIG. **3** is a graphical representation of three non-limiting exemplary types of gearing relationships **48**A, **48**B and **48**C (referred generically as "gearing relationship **48**") for pitch control system **28**. Gearing relationships **48**A-**48**C are plotted as curves on a graph of stabilizer actuation (degrees) versus elevator actuation (degrees) and are superimposed for illustration purposes.

(28) Pitch control system **28** may be configured to permit gearing relationship **48** to be changed based on one or more operating parameters of aircraft **10**. Gearing relationship **48** may be changed based on instructions **40** executable by processor **36**. For example, the changing of gearing relationship **48** may be achieved by way of pitch control system **28** making use of two or more gearing relationships **48** which may be selected and used at different times depending on the operating condition of aircraft **10**. In some embodiments, the desired gearing relationship **48** may be selected by pitch control system **28** based on one or more parameters such as flap angle(s), airspeed and/or Mach number. In some embodiments, gearing relationship(s) **48** may be changed by way of being programmable as a function of one or more parameters such as flap angles, airspeed and/or Mach number.

(29) It is understood that aspects of the present disclosure are not limited to gearing relationships **48**A-**48**C illustrated herein as examples. In various embodiments, gearing relationship **48** may comprise one or more linear portions representative of a constant gear ratio (slope) between stabilizer **24** and elevator **26**. Alternatively or in addition, gearing relationship **48** may comprise one or more non-linear portions representative of a variable gear ratio (slope) between stabilizer **24** and elevator **26**.

(30) By way of example, gearing relationship **48**A may comprise an entirely linear relationship where movement of elevator **26** is proportional to the movement of stabilizer **24** according to a ratio (slope) that is constant over the entire range of motion of stabilizer **24**. On the other hand, gearing relationship **48**B may comprise an entirely non-linear relationship where movement of elevator **26** is geared to the movement of stabilizer **24** according to a ratio (slope) that varies over the range of motion of stabilizer **24**. It is understood that gearing relationship **48** could comprise a combination of one or more linear portions and one or more non-linear portions.

(31) Gearing relationship **48**C may comprise a linear relationship with a dead zone around the neutral position (i.e., 0 degrees) of actuation of stabilizer **24**. For example, gearing relationship **48**C may comprise first portion P**1** over a first range of stabilizer actuation amounts in which a first non-zero stabilizer actuation amount requires a corresponding first non-zero elevator actuation amount. Gearing relationship **48**C may comprise second portion P**2** over a second range of stabilizer actuation amounts in which a second non-zero stabilizer actuation amount requires no elevator actuation amount (i.e., dead zone). Gearing relationship **48**C may also comprise third portion P**3** over a third range of stabilizer actuation amounts in which a third non-zero stabilizer actuation

amount requires a corresponding second non-zero elevator actuation amount. In some embodiments, the second range of stabilizer actuation amounts of second portion P**2** may be between the first range of stabilizer actuation amounts of first portion P**1** and the third range of stabilizer actuation amounts of the third portion P**3**.

(32) The second range of stabilizer actuation amounts of second portion P**2** (dead zone) may comprise a neutral position of stabilizer **24**. The second range of stabilizer actuation amounts may also comprise stabilizer actuation amounts that are near the neutral position and which are typically used for cruise and take-off so that geared actuation of elevator **26** may not be required for one or more phases of flight. For example, having a stabilizer actuation amount that is typically used during cruise within second portion P**2** may help lower cruise drag. Gearing relationship **48**C may be configured to only make use of geared elevator actuation at larger negative stabilizer settings (e.g., lower than −6°) and for positive stabilizer settings that are greater than +2° for example to assist trimming of stabilizer **24** at a maximum speed (i.e., V.sub.FE) of aircraft **10** when flaps are extended. Second portion P**2** may also make more elevator travel available during phases of flight where stabilizer actuation amounts are within second portion P**2**.

(33) In some embodiments, at least one of the first portion P**1**, second portion P**2** and third portion P**3** of gearing relationship **48**C may be linear. In some embodiments, each of the first portion P**1**, second portion P**2** and third portion P**3** of gearing relationship **48**C may be linear. In some embodiments, at least one of the first portion P**1**, second portion P**2** and third portion P**3** of gearing relationship **48**C may be non-linear.

(34) In various embodiments, gearing relationship (e.g., **48**B and **48**C) may comprises a first portion (e.g., P**1**) over a first range of stabilizer actuation amounts and a second portion (e.g., P**2**) over a second range of stabilizer actuation amounts where the first portion and the second portion have different slopes.

(35) FIG. **4** is a schematic illustration of stabilizer **24** and elevator **26** of aircraft **10** of FIG. **1** where stabilizer **24** is actuated to a commanded deflected position represented by line D**1** away from its neutral position represented by line N and elevator **26** is actuated relative to stabilizer **24** according to a corresponding deflected position represented by line D**2**. Line N may be substantially parallel and coincident with a chord of stabilizer **24** when stabilizer **24** is in its neutral position. Line D**1** may be substantially parallel and coincident with the chord of stabilizer **24** when stabilizer **24** is in its actuated/deflected position. Line D**1** may also be substantially parallel and coincident with a chord of elevator **26** when elevator **26** is in its neutral position relative to stabilizer **24**. Line D**2** may be substantially parallel and coincident with the chord of elevator **26** when elevator **26** is in its actuated/deflected position that is electronically geared to the actuated/deflected position of stabilizer **24**. The actuation amount of stabilizer **24** may be represented by a first angular deflection θ**1** (e.g., degrees). The corresponding actuation amount of elevator **26** determined based on gearing relationship **48** may be represented by a second angular deflection θ**2** (e.g., degrees) measured relative to line D**1**.

(36) In various embodiments, gearing relationship **48** may be selected so that the corresponding actuation amount of elevator **26** does not require the entire range of motion of elevator **26**. For example, gearing relationship **48** may be configured to allow for further actuation of elevator **26** in either direction beyond or below angle θ**2** without overly restricting the remaining travel of elevator **26** in order to maintain satisfactorily maneuverability of aircraft **10**. This may still provide a pilot of aircraft **10** the ability to execute pitch commands via input device **46** (see FIG. **2**) intended for elevator **26**. Accordingly, in some embodiments, the actuation amount of elevator **26** that is due to electronic gearing with stabilizer **24** may be considered as the new neutral position of elevator **26** and serve as a new reference point for further actuation/deflection of elevator **26**. Accordingly, in some embodiments of pitch control system **28**, instructions **40** may be configured to cause data processor **36** to reset a neutral position of elevator **26** as the corresponding elevator actuation amount (e.g., D**2**) reached due to electronic gearing relationship **48**. In this arrangement,

input device **46** could remain in its neutral position even after some geared actuation of elevator **26**.
(37) FIG. **5** is a flowchart illustrating method **500** for controlling the pitch of aircraft **10** using movable stabilizer **24** and elevator **26** movably coupled to stabilizer **24**. In some embodiments, method **500** or part(s) thereof may be performed using pitch control system **28** described above. It is understood that functions described above in relation to pitch control system **28** may be incorporated into method **500**. In various embodiments, method **500** may comprise: actuating stabilizer **24** (see block **502**); and actuating elevator **26** based on electronic gearing relationship **48** between stabilizer **24** and elevator **26** (see block **504**). Although blocks **502** and **504** are shown sequentially, it is understood that actuating stabilizer **24** (block **502**) and actuating elevator **26** based on electronic gearing relationship **48** between stabilizer and elevator (block **504**) may be implemented simultaneously.
(38) In some embodiments of method **500**, and as explained above in relation to pitch control system **28**, gearing relationship **48** may comprise first portion P**1** over a first range of stabilizer actuation amounts in which a first non-zero stabilizer actuation amount requires a corresponding first non-zero elevator actuation amount; and second portion P**2** (i.e., dead zone) over a second range of stabilizer actuation amounts in which a second non-zero stabilizer actuation amount requires no elevator actuation amount. In some embodiments, the second range of stabilizer actuation amounts of second portion P**2** comprises a neutral position of stabilizer **24**. In some embodiments, the gearing relationship comprises third portion P**3** over a third range of stabilizer actuation amounts in which a third non-zero stabilizer actuation amount requires a corresponding second non-zero elevator actuation amount. The second range of stabilizer actuation amounts may be between the first range of stabilizer actuation amounts and the third range of stabilizer actuation amounts. In some embodiments, at least one of first portion P**1**, second portion P**2** and third portion P**3** of gearing relationship **48** may be linear. In some embodiments, each of first portion P**1**, second portion P**2** and third portion P**3** of gearing relationship **48** may be linear.
(39) In various embodiments of method **500**, gearing relationship **48** may comprise a first portion over a first range of stabilizer actuation amounts and a second portion over a second range of stabilizer actuation amounts where the first portion and the second portion have different slopes as illustrated in FIG. **3** by gearing relationships **48**B and **48**C. In some embodiments, gearing relationship **48** may comprise a linear portion. In some embodiments, gearing relationship **48** may comprise a non-linear portion.
(40) In some embodiments, method **500** may comprise resetting a neutral position of the elevator to an elevator actuation amount attributed to electronic gearing relationship **48**.
(41) The above description is meant to be exemplary only, and one skilled in the relevant arts will recognize that changes may be made to the embodiments described without departing from the scope of the invention disclosed. For example, the blocks and/or operations in the flowcharts and drawings described herein are for purposes of example only. There may be many variations to these blocks and/or operations without departing from the teachings of the present disclosure. For instance, blocks may be added, deleted, or modified. The present disclosure may be embodied in other specific forms without departing from the subject matter of the claims. Also, one skilled in the relevant arts will appreciate that while the systems and methods disclosed and shown herein may comprise a specific number of elements/components, the systems and methods could be modified to include additional or fewer of such elements/components. The present disclosure is also intended to cover and embrace all suitable changes in technology. Modifications which fall within the scope of the present invention will be apparent to those skilled in the art, in light of a review of this disclosure, and such modifications are intended to fall within the appended claims. Also, the scope of the claims should not be limited by the preferred embodiments set forth in the examples, but should be given the broadest interpretation consistent with the description as a whole.

Claims

1. An aircraft pitch control system comprising: a movable horizontal stabilizer; a first actuator for actuating the horizontal stabilizer; an elevator movably coupled to the horizontal stabilizer; a second actuator for actuating the elevator; and a computer configured to control the second actuator using an electronic gearing relationship between the horizontal stabilizer and the elevator to cause the elevator to be electronically geared to the horizontal stabilizer, wherein: the electronic gearing relationship is changeable based on one or more operating parameters of the aircraft, the one or more operating parameters of the aircraft including a flap angle; the electronic gearing relationship comprises: a first portion over a first range of stabilizer actuation amounts in which a first stabilizer actuation amount requires a corresponding first elevator actuation amount that is smaller than the first stabilizer actuation amount; a second portion defining a dead zone over a non-zero second range of stabilizer actuation amounts in which a non-zero second stabilizer actuation amount requires no corresponding second elevator actuation amount; and a third portion over a third range of stabilizer actuation amounts in which a third stabilizer actuation amount requires a corresponding third elevator actuation amount; and at least one of the first portion and third portion of the gearing relationship is linear.
2. The system as defined in claim 1, wherein the second range of stabilizer actuation amounts of the second portion comprises a neutral position of the stabilizer.
3. The system as defined in claim 1, wherein the second range of stabilizer actuation amounts is between the first range of stabilizer actuation amounts and the third range of stabilizer actuation amounts.
4. The system as defined in claim 1, wherein each of the first portion and third portion of the gearing relationship is linear.
5. The system as defined in claim 1, wherein the electronic gearing relationship comprises a non-linear portion.
6. An aircraft comprising the aircraft pitch control system as defined in claim 1.
7. The system as defined in claim 1, wherein the first portion of the gearing relationship is changeable to be linear and to define a ratio of the first elevator actuation amount to the first stabilizer actuation amount equal to 0.5.
8. The system as defined in claim 1, wherein the third elevator actuation amount is smaller than the third stabilizer actuation amount.